Inference Problem Set

1. estimated standard error: $\frac{s}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$

Test at $\alpha = 0.1$: $z = \frac{52-50}{1} = 2$

compare 2 with 1.645 → $2 > 1.645$ so we reject H_0 at $\alpha = 0.1$

Test at $\alpha = 0.05$ → compare 2 w/ 1.96 → $2 > 1.96$ so we reject H_0 at $\alpha = 0.05$

Test at $\alpha = 0.01$ → compare 2 w/ 2.576 → $2 < 2.576$ so fail to reject H_0 at $\alpha = 0.01$

95% confidence interval for $\mu = \bar{X} \pm 1.96 \cdot SE$

$= 52 \pm 1.96(1) = 52 \pm 1.96$ → (50.04, 53.96)

2. 90% confidence interval: $3.1 \pm 1.645(1.3) = 3.1 \pm 2.1385$

(0.9615, 5.2385) and reject H_0 that $\delta = 0$ at $\alpha = 0.1$ because this interval does not contain 0

95% confidence interval: $3.1 \pm 1.96(1.3) = 3.1 \pm 2.548$

(0.552, 5.648) and reject H_0 that $\delta = 0$ at $\alpha = 0.05$ because this interval does not contain 0

99% confidence interval: $3.1 \pm 2.576(1.3) = 3.1 \pm 3.3488$

(-0.2488, 6.4488) and fail to reject H_0 that $\delta = 0$ at $\alpha = 0.01$ because this interval does contain 0

We reject H_0 at $\alpha = 0.1$ and $\alpha = 0.05$, but fail to reject at $\alpha > 0.01$.

3. 95% confidence interval = $11.5 \pm 1.96\left(\frac{3}{\sqrt{36}}\right) = 11.5 \pm 0.98$

$(10.52, 12.48)$

We fail to reject the H_0 that $\mu = 12$ because our 95% confidence interval contains 12. Thus, there is no evidence to suggest the mean is not 12. Because of this, 12 is certainly a possible value for the population mean.

4. z-statistic for testing equality of means $= \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{s_A^2}{n} + \frac{s_B^2}{n}}} = \frac{25-22}{\sqrt{\frac{8^2}{100} + \frac{7^2}{100}}}$

$= \frac{3}{\sqrt{1.13}} = 2.822$

95% confidence interval for $\mu_A - \mu_B = 3 \pm 1.96\left(\sqrt{1.13}\right) = (0.916, 5.084)$

5. standard error $= \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{0.835} = 0.9138$

test stat $= \frac{14-12}{0.9138} = 2.1886$

6. test stat $= \frac{3.2-2.8}{\sqrt{\underbrace{\frac{1.5^2}{49} + \frac{1.4^2}{64}}_{0.2767}}} = 1.4458$

95% confidence interval: $0.4 \pm 1.96(0.2767) = (-0.142, 0.942)$

At $\alpha = 0.05$, we fail to reject the H_0: $\mu_A - \mu_B = 0$ because $1.4458 < 1.96$ and our 95% confidence interval includes 0. Thus at $\alpha = 0.05$, we have no evidence to suggest the difference in means is not 0. The difference is not statistically large relative to the noise.

Question 7

In [1]:
```
import pandas as pd
import numpy as np

data7 = pd.read_csv('./data/ames_prices.csv')
```

In [2]:
```
data7["Central_Air_Ind"] = (data7["Central.Air"] == "Y").astype(int)
```

Part a

In [3]:
```
data7['Central_Air_Ind'].mean()
```

Out[3]:
```
np.float64(0.9331058020477816)
```

The estimated proportion of homes with central air is 0.933.

Part b

In [4]:
```
np.random.seed(1)
R = 5000
n = len(data7)
bootstrap_estimates = []
for i in range(R):
  boot_sample = data7["Central_Air_Ind"].sample(n=n, replace=True)
  bootstrap_estimates.append(boot_sample.mean())
```

I did 5000 bootstrap estimates and used a seed of 1.

Part c

In [5]:
```
se7 = np.std(bootstrap_estimates,ddof=1)
se7
```

Out[5]:
```
np.float64(0.004574946323265414)
```

I get the bootstrap standard error to be 0.00457.

Part d

In [6]:
```
ci_90 = (data7['Central_Air_Ind'].mean() - 1.645 * se7, data7['Central
```

```
ci_95 = (data7['Central_Air_Ind'].mean() - 1.96 * se7, data7['Central_
ci_99 = (data7['Central_Air_Ind'].mean() - 2.576 * se7, data7['Central
```

In [7]:
```
ci_90, ci_95, ci_99
```

Out[7]:
```
((np.float64(0.92558001534601), np.float64(0.9406315887495532)),
 (np.float64(0.9241389072541814), np.float64(0.9420726968413818)),
 (np.float64(0.9213207403190499), np.float64(0.9448908637765133)))
```

The 90 percent confidence interval is (0.92558001534601, 0.9406315887495532). The 95 percent confidence interval is (0.9241389072541814, 0.9420726968413818). The 99 percent confidence interval is (0.9213207403190499 0.9448908637765133).

Part e

At alpha of 0.1, I reject the null hypothesis that the population proportion of homes with central air is 0.9 because the 90 percent confidence interval does not contain 0.9. At alpha of 0.05, I reject the null hypothesis that the population proportion of homes with central air is 0.9 because the 95 percent confidence interval does not contain 0.9. At alpha of 0.01, I reject the null hypothesis that the population proportion of homes with central air is 0.9 because the 99 percent confidence interval does not contain 0.9.

Question 8

In [8]:
```
np.corrcoef(np.arcsinh(data7['area']), np.arcsinh(data7['price']))
```

Out[8]:
```
array([[1.        , 0.72316615],
       [0.72316615, 1.        ]])
```

The correlation is 0.72316615.

Part a

In [9]:
```
np.random.seed(1)
R = 2000
bootstrap_estimates = []
for i in range(R):
  boot_indices = np.random.choice(n, size=n, replace=True)
  x_boot = data7['area'].iloc[boot_indices]
  y_boot = data7['price'].iloc[boot_indices]
  bootstrap_estimates.append(np.corrcoef(x_boot, y_boot)[0, 1])
```

I used 2000 replications for this and a random seed of 1.

Part b

In [10]:
```
se8 = np.std(bootstrap_estimates, ddof=1)
se8
```

Out[10]:
```
np.float64(0.016502795587168734)
```

I get the standard error to be 0.016502795587168734.

Part c

In [11]:
```
ci_90 = (0.72316615 - 1.645 * se8, 0.72316615 + 1.645 * se8)
ci_95 = (0.72316615 - 1.96 * se8, 0.72316615 + 1.96 * se8)
ci_99 = (0.72316615 - 2.576 * se8, 0.72316615 + 2.576 * se8)
```

In [12]:
```
ci_90, ci_95, ci_99
```

Out[12]:
```
((np.float64(0.6960190512591073), np.float64(0.7503132487408926)),
 (np.float64(0.6908206706491492), np.float64(0.7555116293508507)),
 (np.float64(0.6806549485674533), np.float64(0.7656773514325466)))
```

The 90 percent confidence interval is (0.6960190512591073, 0.7503132487408926). The 95 percent confidence interval is (0.6908206706491492, 0.7555116293508507). The 99 percent confidence interval is (0.6806549485674533,0.7656773514325466).

Part d

At alpha of 0.1, I reject the null hypothesis that the population correlation is 0 because the 90 percent confidence interval does not contain 0. At alpha of 0.05, I reject the null hypothesis that the population correlation is 0 because the 95 percent confidence interval does not contain 0. At alpha of 0.01, I reject the null hypothesis that the population correlation is 0 because the 99 percent confidence interval does not contain 0.

Part e

Area and price must remain paired during resampling so the relationship between the 2 variables can stay in tact. It allows for the relationship to be preserved.

Question 9

In [13]:
```
data9 = pd.read_csv('./data/metabric.csv')
data9.head()
```

Out[13]:

	Age at Diagnosis	Type of Breast Surgery	Cancer Type	Chemotherapy	Hormone Therapy	Lymph nodes examined positive	Muta C
0	43.19	BREAST CONSERVING	Breast Cancer	NO	YES	0.0	
1	48.87	MASTECTOMY	Breast Cancer	YES	YES	1.0	
2	47.68	MASTECTOMY	Breast Cancer	YES	YES	3.0	
3	76.97	MASTECTOMY	Breast Cancer	YES	YES	8.0	
4	78.77	MASTECTOMY	Breast Cancer	NO	YES	0.0	

In [14]:
```
mean9 = data9['Overall Survival (Months)'].mean()
mean9
```

Out[14]:
```
np.float64(129.09429138744105)
```

The mean survival in months is 129.09429138744105.

Part a

In [15]:
```
np.random.seed(1)
n = len(data9)
R = 2000
bootstrap_estimates = []
for i in range(R):
  boot_indices = np.random.choice(n, size=n, replace=True)
  x_boot = data9['Overall Survival (Months)'].iloc[boot_indices]
  bootstrap_estimates.append(np.mean(x_boot))
```

I used 2000 bootstrap replicates and a seed of 1.

Part b

In [16]:
```
se9 = np.std(bootstrap_estimates, ddof=1)
```

```
se9
```

Out[16]:
```
np.float64(2.1864493521763357)
```

The bootstrap standard error is 2.1864493521763357.

Part c

In [17]:
```
ci_90 = (mean9 - 1.645 * se9, mean9 + 1.645 * se9)
ci_95 = (mean9 - 1.96 * se9, mean9 + 1.96 * se9)
ci_99 = (mean9 - 2.576 * se9, mean9 + 2.576 * se9)
```

In [18]:
```
ci_90, ci_95, ci_99
```

Out[18]:
```
((np.float64(125.49758220311098), np.float64(132.69100057177113)),
 (np.float64(124.80885065717543), np.float64(133.37973211770665)),
 (np.float64(123.4619978562348), np.float64(134.7265849186473)))
```

The 90 percent percent confidence interval is (125.49758220311098,132.69100057177113). The 95 percent confidence interval is (124.80885065717543,133.37973211770665). The 99 percent confidence interval is (123.4619978562348,134.7265849186473).

Part d

At alpha of 0.1, I reject the null hypothesis that the population mean is 120 months because the 90 percent confidence interval does not contain 120. At alpha of 0.05, I reject the null hypothesis that the population mean is 120 months because the 95 percent confidence interval does not contain 120. At alpha of 0.01, I reject the null hypothesis that the population mean is 120 months because the 99 percent confidence interval does not contain 120.

Part e

In interpreting the mean estimate of 129.09429138744105, this means that we expect patients to live 129.09429138744105 months, or 10.757869405053090972 years. In interpreting the 90 percent confidence interval, this means we are 90 percent confident the true amount of months (or years) survived is between 125.49758220311098 months (or 10.458143311225658323 years) and 132.69100057177113 (or 11.057595498880521845 years). In interpreting the 95 percent confidence interval, this means we are 95 percent confident the true amount of months (or

years) survived is between 124.80885065717543 months (or 10.400748952833170691 years) and 133.37973211770665 (or 11.114989857273007701 years). In interpreting the 99 percent confidence interval, this means we are 99 percent confident the true amount of months (or years) survived is between 123.4619978562348 months (or 10.2885110964211961 years) and 134.7265849186473 (or 11.227227713684984067 years). These results do not estimate unrestricted mean lifetime because if patients survived longer than the study, then we do not truly know how long they survived, which is why the results do not estimate unrestricted mean lifetime. We are underestimating the survival time because of this.

Question 10

In [21]:
```
data10 = pd.read_csv('./data/nhanes_data_17_18.csv')
data10
```

```
/var/folders/n7/hg2001tn41z7vw3b9b65gyq00000gp/T/ipykernel_90287/3096201681.py:1: DtypeWarning: Columns (0: ServedActiveDutyInUsArmedForces) have mixed types. Specify dtype option on import or set low_memory=False.
  data10 = pd.read_csv('./data/nhanes_data_17_18.csv')
```

Out[21]:

	SEQN	GeneralHealthCondition	EverBreastfedOrFedBreastmilk	AgeSto
0	93703.0	NaN	1.0	
1	93704.0	NaN	1.0	
2	93705.0	Good	NaN	
3	93706.0	Very good	NaN	
4	93707.0	Good	NaN	
...	...	...	...	
8361	102952.0	Very good	NaN	
8362	102953.0	Fair or	NaN	
8363	102954.0	Good	NaN	
8364	102955.0	Very good	NaN	
8365	102956.0	Good	NaN	

8366 rows × 198 columns

In [22]:
```
data10 = data10[(data10["AgeInYearsAtScreening"] >= 18) & (data10["Sle
```

```
data10
```

Out[22]:

	SEQN	GeneralHealthCondition	EverBreastfedOrFedBreastmilk	AgeSto
2	93705.0	Good	NaN	
3	93706.0	Very good	NaN	
5	93708.0	Good	NaN	
6	93709.0	NaN	NaN	
7	93711.0	Very good	NaN	
...	...	...	...	
8359	102949.0	Excellent	NaN	
8361	102952.0	Very good	NaN	
8362	102953.0	Fair or	NaN	
8363	102954.0	Good	NaN	
8365	102956.0	Good	NaN	

5489 rows × 198 columns

In [26]:
```
mean10 = data10['SleepHoursWeekdaysOrWorkdays'].mean()
mean10
```

Out[26]:
```
np.float64(7.62124248496994)
```

The mean number sleep is 7.62124248496994 hours.

Part a

In [24]:
```
np.random.seed(1)
n = len(data10)
R = 2000
bootstrap_estimates = []
for i in range(R):
  boot_indices = np.random.choice(n, size=n, replace=True)
  x_boot = data10['SleepHoursWeekdaysOrWorkdays'].iloc[boot_indices]
  bootstrap_estimates.append(np.mean(x_boot))
```

I used a random seed of 1 and 2000 replicates.

Part b

In [25]:
```
se10 = np.std(bootstrap_estimates, ddof=1)
se10
```

Out[25]:
```
np.float64(0.02309249977636107)
```

The bootstrap standard error is 0.02309249977636107.

Part c

In [27]:
```
ci_90 = (mean10 - 1.645 * se10, mean10 + 1.645 * se10)
ci_95 = (mean10 - 1.96 * se10, mean10 + 1.96 * se10)
ci_99 = (mean10 - 2.576 * se10, mean10 + 2.576 * se10)
```

In [28]:
```
ci_90, ci_95, ci_99
```

Out[28]:
```
((np.float64(7.583255322837826), np.float64(7.6592296471020545)),
 (np.float64(7.575981185408272), np.float64(7.666503784531608)),
 (np.float64(7.561756205546034), np.float64(7.680728764393846)))
```

The 90 percent percent confidence interval is (7.583255322837826,7.6592296471020545). The 95 percent confidence interval is (7.575981185408272,7.666503784531608). The 99 percent confidence interval is (7.561756205546034,7.680728764393846).

Part d

At alpha of 0.1, I reject the null hypothesis that the population mean is 8 hours because the 90 percent confidence interval does not contain 8. At alpha of 0.05, I reject the null hypothesis that the population mean is 8 hours because the 95 percent confidence interval does not contain 8. At alpha of 0.01, I reject the null hypothesis that the population mean is 8 hours because the 99 percent confidence interval does not contain 8.

Part e

Since we reject the null hypothesis at all confidence levels, we can say that there is evidence to suggest that the population mean of hours slept is not 8. The statistical significance and practical significance do not tell same story though because while statistically we say that it is not 8 hours, practically this is the difference of just under 30 minutes. The practical difference comes in whether a little under 30 minutes extra sleep is truly useful or not, which doctors would have to be consulted for to know that.